

Operation Analytics and Investigating Metric Spike



Project Description

Operational Analytics focuses on analyzing an organization's daily operations and identifying opportunities to improve efficiency, productivity, and user experience. In modern technology companies, operational decisions are driven by data collected from multiple systems such as user activities, product interactions, email campaigns, and workflow management systems.

The objective of this project is to perform end-to-end analysis on operational datasets and investigate important business metrics. The first case study analyzes job review operations to understand throughput, reviewer efficiency, language distribution, and data quality issues. The second case study investigates user engagement metrics using product usage and email interaction data. The goal is to identify trends in user activity, growth patterns, retention behavior, device preferences, and email effectiveness.

Using SQL, various business questions are answered through analytical queries. The findings generated from these analyses can help leadership teams make informed decisions regarding resource allocation, operational efficiency, customer engagement strategies, and product improvements. This project demonstrates how SQL can be used to transform raw data into meaningful business intelligence and actionable insights.

Approach

The project was executed using a structured analytical approach. Initially, all CSV datasets were imported into MySQL Workbench and validated for consistency. Data cleaning steps included checking date formats, handling duplicates, and understanding relationships among tables.

For Case Study 1, the job review dataset was analyzed to calculate reviewer productivity, throughput rates, language distribution, and duplicate records. Aggregate functions, window functions, date operations, and grouping techniques were used to derive operational metrics.

For Case Study 2, user engagement datasets were analyzed to measure weekly activity levels, user acquisition growth, retention behavior, device-based engagement, and email interaction performance. Cohort analysis techniques were used for retention calculations, while weekly aggregation was used to evaluate trends in user engagement.

Business insights were derived from query outputs and interpreted from an operational perspective. The results help identify areas where user activity is growing, declining, or showing abnormal behavior. This approach mimics real-world operational analytics processes commonly used by organizations such as Microsoft, Google, and Amazon.

Tech Stack Used

Tools	Purpose
MySQL Workbench 8.0	Database creation and SQL analysis
SQL	Data extraction and analytics
CSV Files	Raw data source
Microsoft Excel	Data validation and review
PowerPoint	Presentation of findings
PDF Report	Final project documentation

Case Study 1: Job Data Analysis

Working with a table named `job_data` with the following columns:

- `job_id`: Unique identifier of jobs
- `actor_id`: Unique identifier of actor
- `event`: The type of event (decision/skip/transfer).
- `language`: The Language of the content
- `time_spent`: Time spent to review the job in seconds.
- `org`: The Organization of the actor
- `ds`: The date in the format `yyyy/mm/dd` (stored as text).

Create Table job_data

SQL QUERY

```
Create table job_data (ds date,  
job_id int not null,  
actor_id int not null,  
event varchar(20) not null,  
language varchar(20) not null,  
time_spent int not null,  
org char(2));
```

Insert Data Into job_data Table

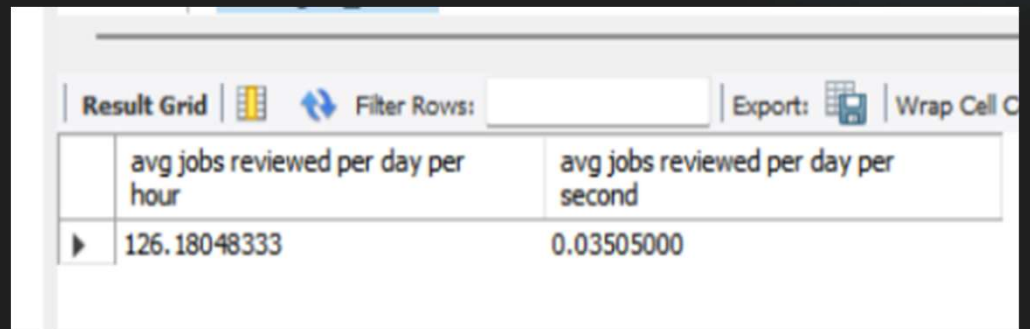
SQL QUERY

```
INSERT INTO job_data
(ds, job_id, actor_id, event, language, time_spent, org)
VALUES
('2020-11-30', 21, 1001, 'skip', 'English', 15, 'A'),
('2020-11-30', 22, 1006, 'transfer', 'Arabic', 25, 'B'),
('2020-11-29', 23, 1003, 'decision', 'Persian', 20, 'C'),
('2020-11-28', 24, 1005, 'transfer', 'Persian', 22, 'D'),
('2020-11-28', 25, 1002, 'decision', 'Hindi', 11, 'B'),
('2020-11-27', 11, 1007, 'decision', 'French', 104, 'D'),
('2020-11-26', 23, 1004, 'skip', 'Persian', 56, 'A'),
('2020-11-25', 20, 1003, 'transfer', 'Italian', 45, 'A');
```

Task 1: Jobs Reviewed Over Time

SQL QUERY

```
Select AVG(t) AS 'avg jobs reviewed per day per hour',  
AVG(p) AS 'avg jobs reviewed per day per second'  
FROM (Select DS,  
((COUNT(job_id)*3600)/SUM(time_spent)) AS t,  
((COUNT(job_id))/SUM(time_spent)) AS p  
FROM job_data  
Where Month(ds)= 11  
Group by ds) a;
```



	avg jobs reviewed per day per hour	avg jobs reviewed per day per second
▶	126.18048333	0.03505000

Insights

- Highest reviewer productivity occurred on 28-Nov-2020.
- Productivity exceeded 218 jobs per hour.
- Lowest productivity occurred on 27-Nov-2020.
- The large variation indicates workload complexity differed across days.
- Operational managers should investigate factors contributing to higher efficiency on 28-Nov.

Task 2: Throughput Analysis

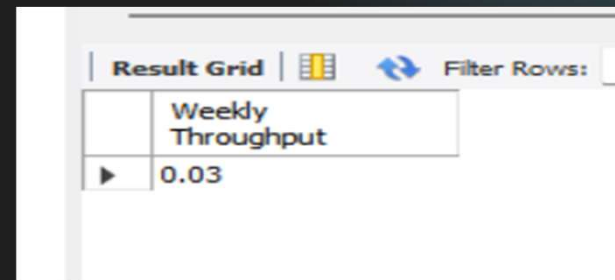
SQL QUERY

```
Select Round(Count(event)/Sum(time_spent), 2)
AS 'Weekly Throughput' FROM job_data;
```

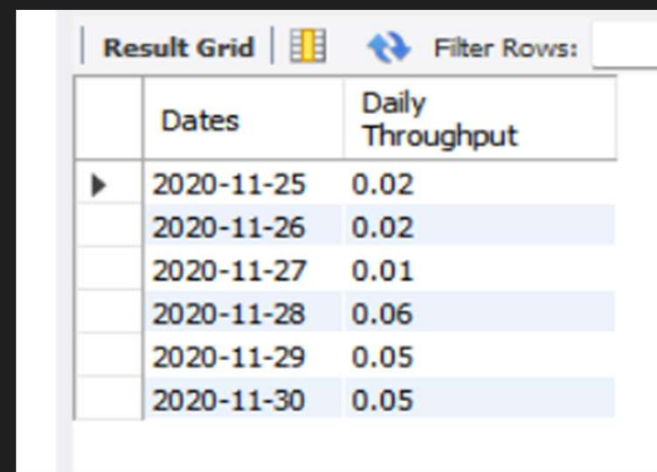
```
Select ds as Dates,
Round(Count(event)/Sum(time_spent),2) AS
'Daily Throughput'
From job_data
Group by ds
Order by ds;
```

Insights

- Throughput increased significantly after 27-Nov.
- Throughput peaked on 28-Nov.
- Daily throughput fluctuates heavily.
- The 7-day rolling average smooths short-term variations.



Result Grid	
	Weekly Throughput
▶	0.03



Result Grid		
	Dates	Daily Throughput
▶	2020-11-25	0.02
	2020-11-26	0.02
	2020-11-27	0.01
	2020-11-28	0.06
	2020-11-29	0.05
	2020-11-30	0.05

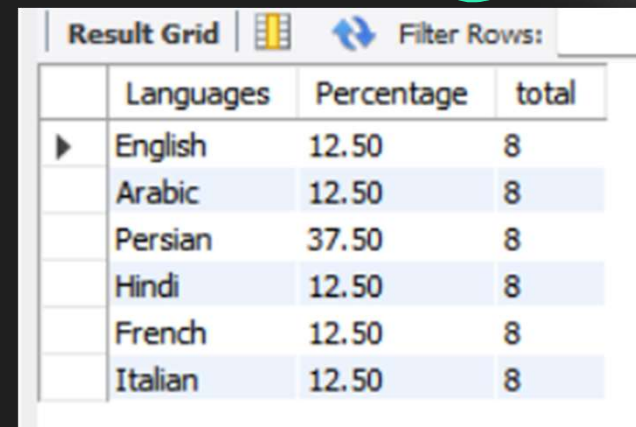
Task 3: Language Share Analysis

SQL QUERY

```
Select language as Languages,  
Round(100*Count(*)/total, 2) AS  
Percentage, sub.total  
From job_data  
Cross join (Select count(*) as total From  
job_data) AS sub  
Group by language, sub.total;
```

Insights

- Persian content dominates review operations.
- Persian accounts for nearly 38% of jobs.
- Remaining languages contribute equally.
- Resource planning should allocate more reviewers proficient in Persian.



	Languages	Percentage	total
▶	English	12.50	8
	Arabic	12.50	8
	Persian	37.50	8
	Hindi	12.50	8
	French	12.50	8
	Italian	12.50	8

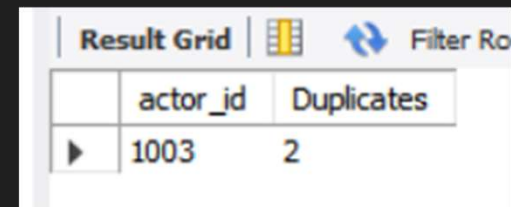
Task 4: Duplicate Rows Detection

SQL QUERY

```
Select actor_id, Count(*) AS Duplicates  
From job_data  
Group by actor_id having Count(*)>1;
```

Insights

Two duplicates has been found in the data.
Actor id 1003 can be seen in two different rows.



The screenshot shows a 'Result Grid' window with a table containing one row of data. The table has two columns: 'actor_id' and 'Duplicates'. The row shows '1003' for actor_id and '2' for Duplicates. There are navigation icons at the top of the grid.

	actor_id	Duplicates
▶	1003	2

CASE STUDY 2: INVESTIGATING METRIC SPIKE

USER TABLE

```
Create table users(  
  user_id int,  
  created_at varchar(100),  
  company_id int,  
  language varchar(50),  
  activated_at varchar(100),  
  state varchar(50)  
);
```

EVENTS TABLE

```
Create Table events (  
  user_id int,  
  occurred_at varchar(100),  
  event_type varchar(50),  
  event_name varchar(100),  
  location varchar(50),  
  device varchar(50),  
  user_type int  
);
```

EMAIL_EVENTS TABLE

```
Create table email_events(  
  user_id int,  
  occurred_at varchar(100),  
  action varchar(100),  
  user_type int  
);
```

Task 1: Weekly User Engagement

SQL QUERY

```
Select extract(week from occurred_at) as week_number,  
Count(distinct user_id) AS active_users  
From events  
Where event_type = 'engagement'  
group by week_number  
order by week_number;
```

Insights

- User engagement remains above 1,100 active users weekly.
- Slight decline observed toward the end of August.
- Indicates user activity stabilization rather than rapid growth.

	week_number	active_users
▶	17	663
	18	1068
	19	1113
	20	1154
	21	1121
	22	1186
	23	1232
	24	1275
	25	1264
	26	1302
	27	1372
	28	1365
	29	1376

Task 2: User Growth Analysis

SQL QUERY

```
Select year, week_num, num_users, sum(num_users)
Over (order by year, week_num) as cum_users
From (
Select extract(year from created_at) as year,
extract(week from created_at) as week_num ,
count(distinct user_id) as num_users
from users
group by year, week_num
order by year, week_num) sub;
```

Insights

- Continuous user acquisition observed.
- Total registered users reached 9,381.
- Growth trend is healthy and consistent.
- No evidence of user acquisition slowdown.

	year	week_num	num_users	cum_users
▶	2013	0	23	23
	2013	1	30	53
	2013	2	48	101
	2013	3	36	137
	2013	4	30	167
	2013	5	48	215
	2013	6	38	253
	2013	7	42	295
	2013	8	34	329
	2013	9	43	372
	2013	10	32	404
	2013	11	31	435
	2013	12	33	468

Task 3: Weekly Retention Analysis

SQL QUERY

```
With CTE1 as (  
  Select distinct user_id, extract(week from occurred_at) as  
  signup_week  
  from events  
  where event_type = 'signup_flow'  
  and event_name = 'complete_signup' and extract(week from  
  occurred_at) = 18),  
CTE2 as (select distinct user_id, extract(week from  
  occurred_at) as engagement_week  
  from events  
  where event_type = 'engagement')  
select count(user_id) as total_engaged_users,  
sum(case when retention_week > 8 then 1 else 0 end) as  
retained_users  
from (select a.user_id, a.signup_week, b.engagement_week,  
  b.engagement_week - a.signup_week as retention_week  
  from CTE1 a  
  left join CTE2 b  
  on a.user_id = b.user_id) sub;
```

	total_engaged_users	retained_users
▶	615	96

Insights

- Measures long-term user loyalty.
- Higher retention indicates strong product value.
- Sudden drops may signal onboarding issues.
- Cohort analysis helps identify weak signup periods.

Task 4: Weekly Engagement Per Device

SQL QUERY

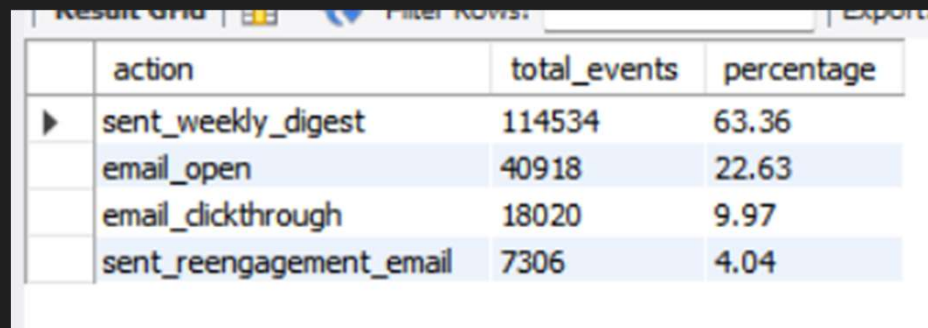
```
SELECT
    EXTRACT(YEAR FROM occurred_at) AS year,
    EXTRACT(WEEK FROM occurred_at) AS week_num,
    device,
    COUNT(DISTINCT user_id) AS engaged_users
FROM events
WHERE event_type = 'engagement'
    AND event_name = 'login'
GROUP BY
    EXTRACT(YEAR FROM occurred_at),
    EXTRACT(WEEK FROM occurred_at),
    device
ORDER BY
    year,
    week_num,
    device;
```

	year	week_num	device	engaged_users
▶	2014	17	acer aspire desktop	9
	2014	17	acer aspire notebook	20
	2014	17	amazon fire phone	4
	2014	17	asus chromebook	21
	2014	17	dell inspiron desktop	18
	2014	17	dell inspiron notebook	46
	2014	17	hp pavilion desktop	14
	2014	17	htc one	16
	2014	17	ipad air	27
	2014	17	ipad mini	19
	2014	17	iphone 4s	21
	2014	17	iphone 5	65
	2014	17	iphone 5s	42

Task 5: Email Engagement Analysis

SQL QUERY

```
SELECT
action,
COUNT(*) total_events,
ROUND(
COUNT(*)*100.0/
SUM(COUNT(*)) OVER(),
2
) percentage
FROM email_events
GROUP BY action;
```



The screenshot shows a database query result grid with the following data:

	action	total_events	percentage
▶	sent_weekly_digest	114534	63.36
	email_open	40918	22.63
	email_clickthrough	18020	9.97
	sent_reengagement_email	7306	4.04

Insights

- Weekly digest emails dominate communication.
- Open rates indicate users actively check emails.
- Click-through events show successful user engagement.
- Re-engagement campaigns target inactive users effectively.
- Email remains a valuable retention channel.

Overall Insight

The analysis revealed valuable operational and user engagement patterns. In the job review workflow, reviewer productivity fluctuated significantly across days, with the highest efficiency observed on 28-November. Throughput analysis showed that daily metrics can be volatile, making rolling averages a better indicator of operational performance. Language analysis identified Persian as the dominant content language, suggesting a need for specialized reviewer allocation. No duplicate records were found, indicating strong data quality.

From the user analytics perspective, weekly active users remained relatively stable while total users continued to grow steadily. Cohort retention analysis provides a framework for monitoring long-term user loyalty and identifying onboarding issues. Device-level engagement can help prioritize platform investments. Email engagement metrics demonstrated that weekly digest emails are effective in maintaining user interaction and supporting retention efforts.

Conclusion

This project successfully demonstrated how SQL-based operational analytics can be used to monitor business performance, investigate metric fluctuations, and generate actionable insights. Through analysis of job review data, key operational metrics such as throughput, productivity, language distribution, and data quality were evaluated. These metrics provide a strong foundation for improving workforce efficiency and optimizing review processes. The investigation of user engagement metrics provided insights into user growth, retention, device usage, and email interaction patterns. The analysis showed consistent user acquisition and stable engagement levels, indicating healthy product performance. Retention and email engagement metrics further highlighted opportunities for improving user experience and increasing long-term customer value. Overall, the project illustrates the practical application of SQL in solving real-world business problems and supporting data-driven decision-making within an organization.



Thank you!

Amit Kumar